QISKIT FALL FEST 2026 · MEMORIAL UNIVERSITY

Lights On, Oil Off

Does Newfoundland need a new oil-burning turbine to retire Holyrood?

Unsupervised Engineers

Sai Ganesh · Gideon Clottey

Theme: Quantum for a Sustainable Newfoundland and Labrador

THE PROBLEM

Replace Holyrood's oil, keep the lights on

- Holyrood burns heavy fuel oil; it retires around 2030
- Lifeline: the Labrador-Island Link, one 700 MW line
- Hydro's plan: Unit 8 (hydro) + a diesel turbine, 150 MW each
- The alternative people raise: wind and heat pumps
- Winter risk: DarkNL (Jan 2014), 190,000 customers lost power
- Jan 2026: frazil ice (slush in the intakes) shut Bay d'Espoir

INTENDED BENEFIT

Retire the oil without building new diesel: less CO2, and a clear test for a CAD 1 billion decision

~600 kt

CO2 a year from Holyrood's oil (2022 to 2024)

CAD 1.0 billion

for the proposed 150 MW diesel turbine

2.8 hours

NL Hydro's limit on expected blackout a year

Sources: NL GHG facility data 2016-22 and ECCC large-facility data (2026); NL Hydro, Avalon Combustion Turbine application, Rev. 1 (PUB, Apr 2026); NL Hydro, 2024 Resource Adequacy Plan, Rev. 2 (PUB, Aug 2024); NL Hydro DarkNL briefing to the PUB (Jan 2014) and January 2026 power warning notices.

THE GOAL

One question, one rule

Can wind and heat pumps do the new turbine's job and keep expected blackouts under 2.8 hours a year?

INPUT

- Hourly island demand, 2022-26
- 46 winters of weather
- Plant outage odds (NL Hydro)

OUTPUT

- Blackout hours, cost and CO2 for 4,096 plans (12 yes/no options)
- The link outage rate where the turbine is needed

USE

- Evidence for the 2030 build decision before the regulator (PUB)

BASELINE

- Monte Carlo (random simulation)
- Tuned importance sampling
- Exact counting as the answer key

METRIC

- Blackout hours vs the 2.8 h rule
- Runs needed for a confident pass or fail

Scope: the island grid in winter 2030, supply versus demand only; small enough to check against exact answers.

Rule: NL Hydro, 2024 Resource Adequacy Plan, Rev. 2 (PUB, Aug 2024), expected loss of load at most 2.8 hours a year. Fields follow Step 2 of the Qiskit Fall Fest 2026 rule book.

THE PIPELINE

From weather to a yes-or-no answer

1

Public data

Hourly island demand

46 winters of weather

Plant outage odds

→

2

AI model

Weather in, demand out

Gradient-boosted trees

Extra rule for record cold

→

3

Grid model

Plants, link, wind, heat pumps

Every hour of 46 winters

Checked vs NL Hydro

→

4

Risk per plan

Chance an hour is short of power

Qiskit amplitude estimation

Baseline: Monte Carlo

→

5

Decision

4,096 plans scored

Pass: 2.8 h or less

When is the turbine needed?

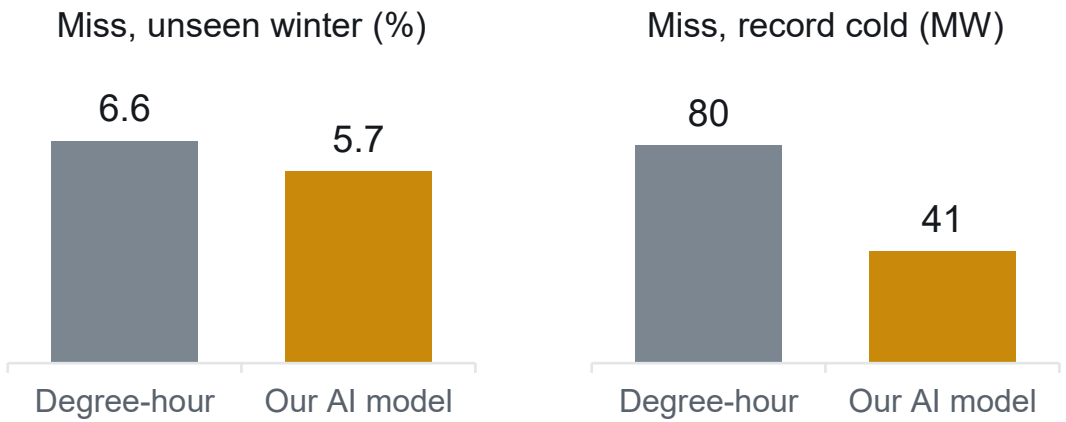
Simulator: Qiskit Aer, noise-free and with an IBM noise model
Hardware: real IBM chip (ibm_quebec, 156 qubits): 22 circuits, 64 s
Proof of concept: notebook with saved outputs, plus an interactive demo

- Classical
- AI
- Quantum (Qiskit)

Data: Statistics Canada / CCEI High-Frequency Electricity Data (NL island demand); ERA5 reanalysis (Hersbach et al. 2020) via Open-Meteo; NL Hydro filings to the PUB (2024-26).
Code: Qiskit 2.6, Qiskit Aer 0.18, scikit-learn 1.9.

Learning island demand from the weather

- Predicts **island demand (MW) for every hour** from the weather
- Gradient-boosted trees, trained on **23,362 hours** (Sep 2022 to Aug 2025)
- Inputs: temperature now and earlier, wind, sun, snow in 3 cities; hour, weekday, holidays
- **Record cold:** trees go flat past the coldest hour they saw (−16.8 °C), so below it we add 29 MW per degree colder
- Tested on the unseen 2025-26 winter (**7,502 hours**)



Miss: average gap from real hourly demand; lower is better
Record cold: hid all 757 hours below −8 °C, then predicted them
Degree-hour: the utility standard, a straight line in temperature

OUR FORECAST VS NL HYDRO'S, WINTER 2026-27

1.63 h ours 1.8 h NL Hydro

Expected blackout hours this winter; within about 10%

Statistics Canada / CCEI HFED (NL island demand, Sep 2022 to Nov 2026); ERA5 via Open-Meteo; scikit-learn (Pedregosa et al. 2011); NL Hydro, 2025 Near-Term Reliability Report (PUB, Nov 2025) for the 1.8 h figure.

12 yes-or-no choices, 4,096 plans

	Hydro's plan	Cheapest, no turbine	Strongest, no turbine	Strongest + turbine
SUPPLY				
Diesel turbine at Holyrood, 150 MW	●			●
Bay d'Espoir Unit 8 (hydro), 150 MW	●	●	●	●
Wind, Avalon, 100 MW		●	●	●
Wind, Burin, 100 MW		●	●	●
Wind, west coast, 100 MW			●	●
Battery, 25 MW for 4 hours			●	●
Battery, 50 MW for 4 hours			●	●
DEMAND				
Heat pumps, 15,000 electric-heated homes		●	●	●
Heat pumps, 35,000 electric-heated homes		●	●	●
Heat pumps, 10,000 oil-heated homes		●		
Heat pumps, 30,000 oil-heated homes		●		
Smart thermostats, 40 MW off the peak		●	●	●
Blackout hours a year, link down 1.1%	**1.91**	**1.75**	**0.92**	**0.20**

EACH PLAN IS SCORED ON

- Blackout hours: replay 46 winters, hour by hour, at 2030 demand
- Yearly cost: build and run, minus savings on power bills and oil
- CO2: oil burned or avoided
- All 4,096 plans scored on a classical computer

HEAT PUMPS CUT TWO WAYS

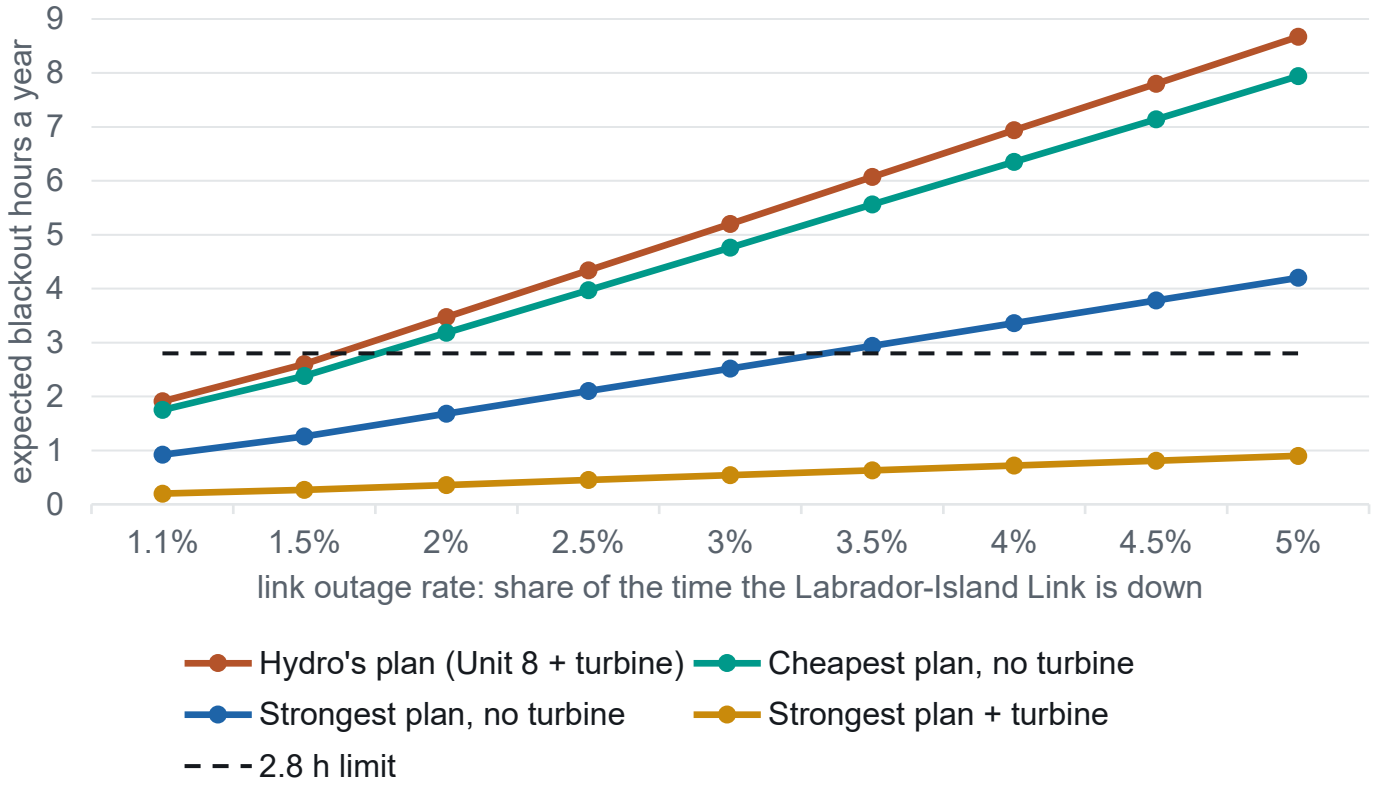
Electric-heated homes: lower winter peak. Oil-heated homes: less CO2, but more load on the grid.

Options: gridrisk/portfolio.py. Turbine CAD 995.9M and Unit 8 CAD 1.08B: NL Hydro filings to the PUB (2026). Battery cost and capacity value: NL Hydro battery and ELCC studies. Other costs are our stated assumptions.

CLASSICAL MODEL · ALL 4,096 PLANS

Turbine needed only if the link gets worse

How to read: each line is a plan. Further right, the link is down more often. Higher, more blackout hours. Below the dashed line, the plan passes.



Today: no turbine

The link was down 1.1% of the past 12 months. The cheapest plan without the turbine passes: 1.75 h a year

Cost: about a tie

Yearly cost after bill savings, cheapest passing plan: CAD 66.6M without the turbine, 64.4M with it. Without it: 10 kt less CO2

Break-even ~3%

Down more than 3.3% of the time, even our strongest no-turbine plan fails. NL Hydro plans for 5%

Our grid model, all 4,096 plans. Link outage rate 1.1%: NL Hydro, Q3 2026 Rolling 12-Month Asset Performance (PUB, Nov 2026); 5% planning value: 2025 Near-Term Reliability Report. Costs: Avalon CT application Rev. 1 (Apr 2026), Unit 8 update (Oct 2026), our assumptions for the rest.

Quantum only where classical has to simulate

TASK	CAN A NORMAL COMPUTER DO IT EXACTLY?	OUR CHOICE
Risk in one winter hour	Yes: a fast exact formula	Classical. It is our answer key
Risk through a multi-day cold snap	Not at scale: the work doubles with every unit (256 states at 6 units, about 9 billion at 31), so planners simulate with Monte Carlo	**Quantum: amplitude estimation replaces that simulation**
Finding the best of 4,096 plans	Yes: score every plan	Classical. A quantum search was slower in 41 of 45 tests

WHY THE COLD SNAP SUITS AMPLITUDE ESTIMATION

1 in 1,000
a plan's risk: about 1 winter hour in 1,000

Hard line
close calls near 2.8 h need that tiny number within a few percent

4× vs 2×
runs to halve the error: Monte Carlo vs amplitude estimation

Honest limit: a 6-unit, 7-day snap needs 64 qubits and 16,690 two-qubit gates; today's chips manage about 30. This is for future, error-corrected machines.

Exact state counts: experiments/e6_cold_snap.py (best exact method: 4 × 2^units states). Quantum search test: Layden et al. (2023) method, notebook section 13. NL Hydro January 2026 power warning notices (Bay d'Espoir frazil ice).

SIMULATOR · QISKIT AER, NOISE-FREE

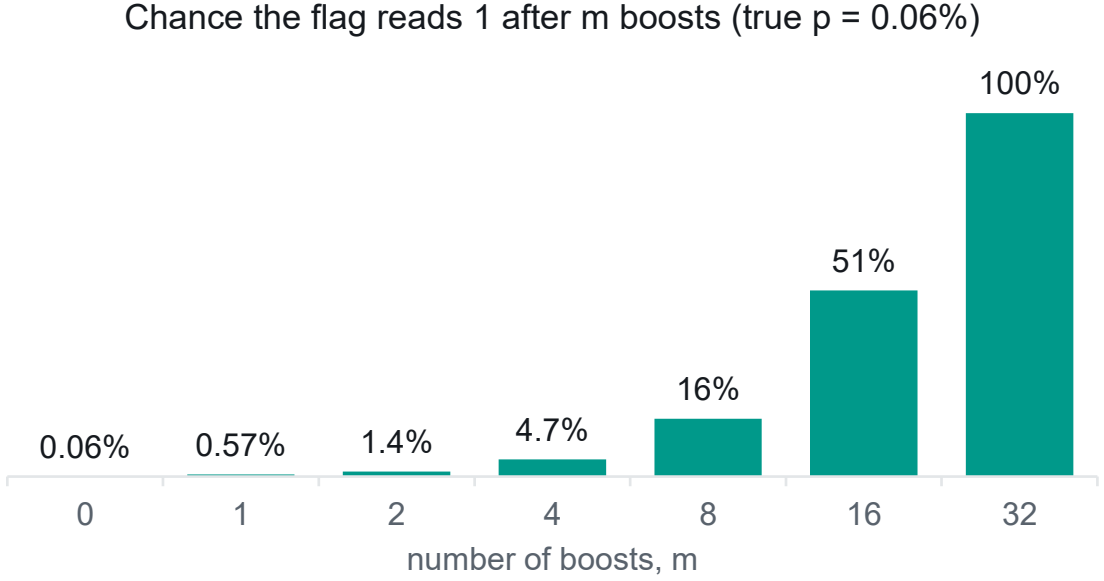
Make the rare event common, then measure it

MONTE CARLO (CLASSICAL)

- Simulate random winter hours, count the ones short of power
- 13,300 runs: only about 8 are short
- Counting 8 events: **error about 35%**

AMPLITUDE ESTIMATION (QUANTUM)

- One circuit holds every hour at once; a flag qubit is 1 with chance p
- Each boost (a Grover step) raises that chance along a known curve
- Measure after 0 to 32 boosts, fit the curve: that gives p
- Same 13,300 runs: **error 4.4%**



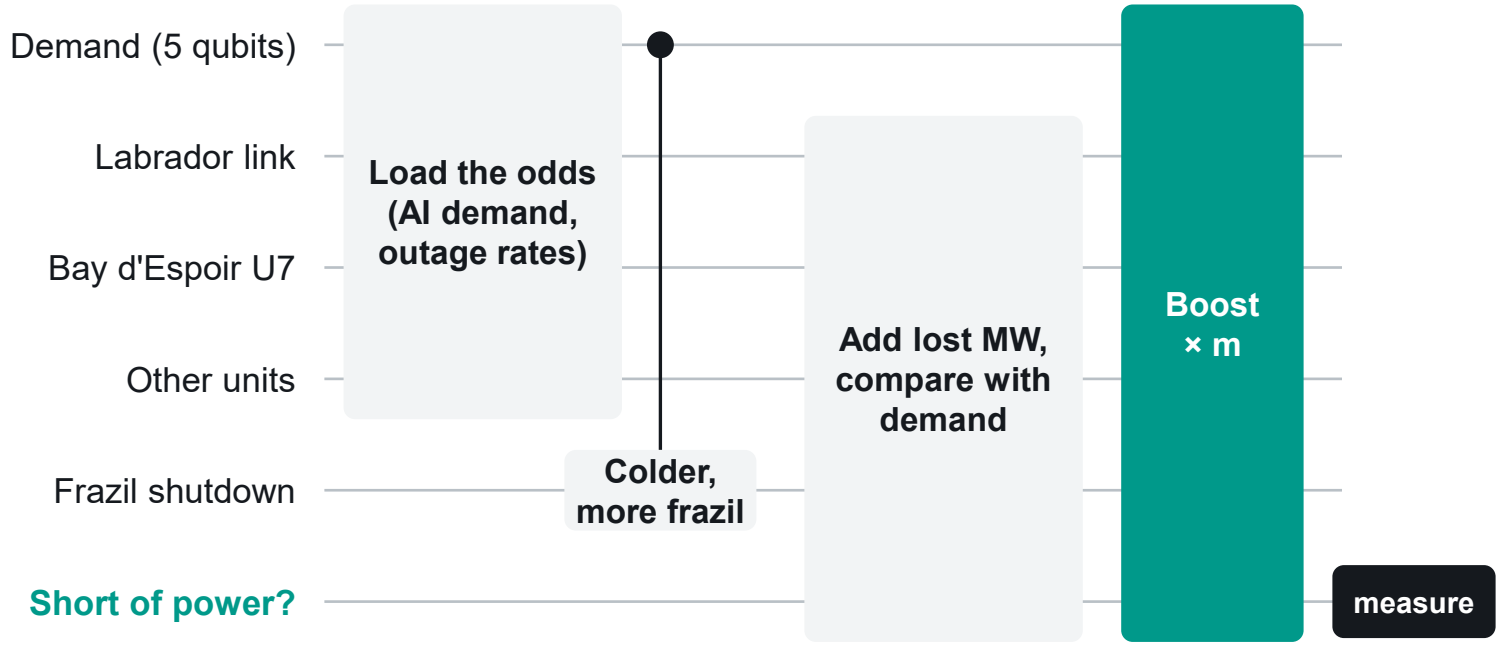
The rule: chance after m boosts = $\sin^2((2m + 1)\theta)$, with $p = \sin^2\theta$. A tiny angle, added up many times, becomes big and easy to measure.

To halve the error: Monte Carlo needs 4× the runs, amplitude estimation only 2×

Example: experiments/e3_quantum_core.py, today's grid, one winter hour, 24-qubit circuit on Qiskit Aer (noise-free), 300 tries. A run is one use of the grid model; one measurement after m boosts counts as 2m + 1 runs. Brassard et al. (2002); Suzuki et al. (2020).

QISKIT

One circuit holds every outage at once



1. **Load the odds:** every mix of demand level and outages, each at its real chance
2. **Colder, more frazil:** the shutdown gets likelier on cold hours
3. **Add lost MW:** the flag qubit turns to 1 if supply falls short
4. **Boost, measure, fit:** gives p; p × 2,904 winter hours = blackout hours a year

24 qubits

today's grid, one winter hour: matches exact counting

22 qubits

a 3-day cold snap: about 3,500 two-qubit gates

3 qubits

a small exact version of a real case, run on the IBM chip

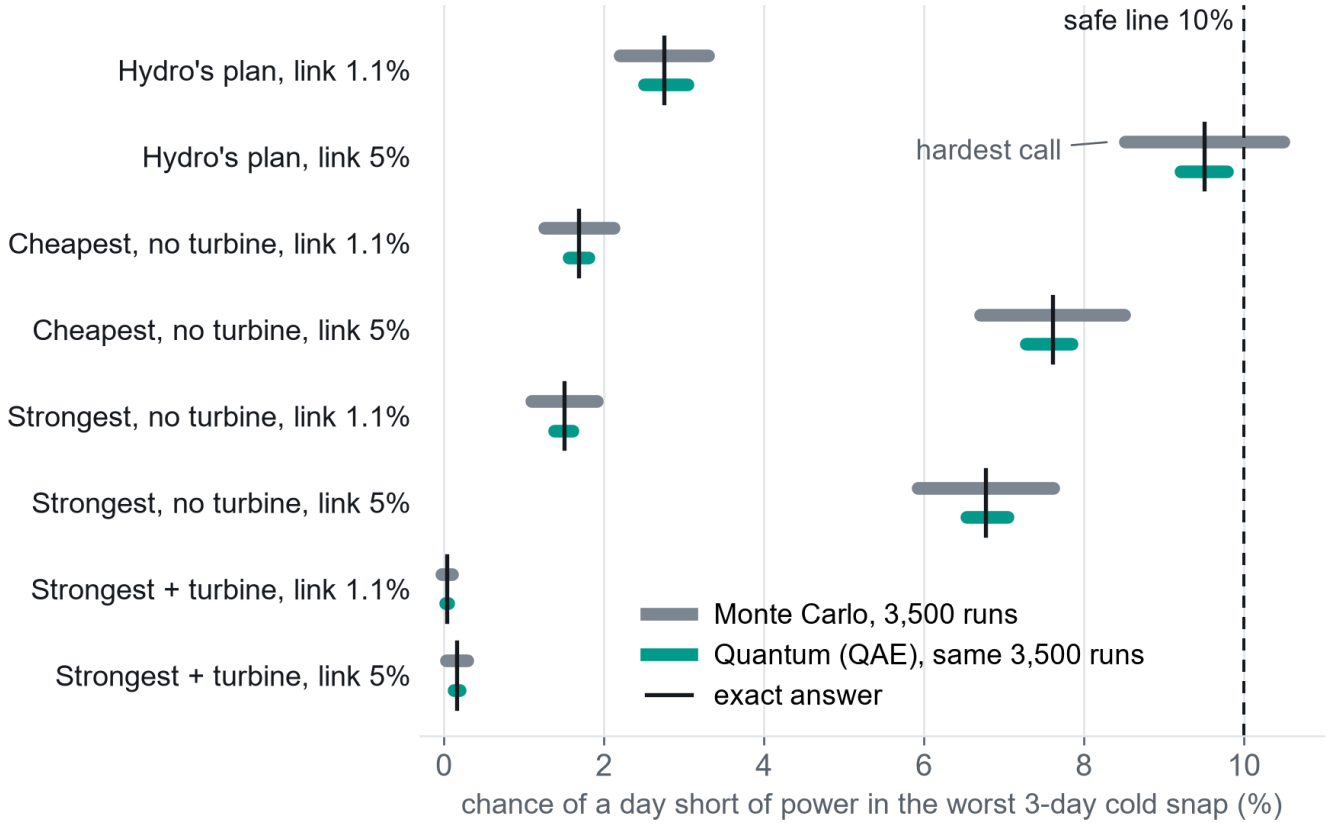
Qubit: n of them hold 2^n cases at once.

Two-qubit gate: a step that links two qubits; the main source of errors on real chips, so the count measures circuit size.

Built with Qiskit 2.6 and Qiskit Aer 0.18. Maximum-likelihood amplitude estimation: Suzuki et al. (2020). Circuit layout after Woerner and Egger (2019). Exact enumeration of the same model is the ground truth.

SIMULATOR · QISKIT AER, NOISE-FREE

Same runs, a sharper answer



How to read: tick = exact answer. Bars = where 95% of 200 tries land. Shorter is better.

THE TEST

4 plans × 2 link rates through the worst 3-day cold snap. Safe = under a 10% chance of a day short of power.

8 of 8
quantum estimates land on the exact answer

2 to 3.5× smaller error
than Monte Carlo, with the same 3,500 runs

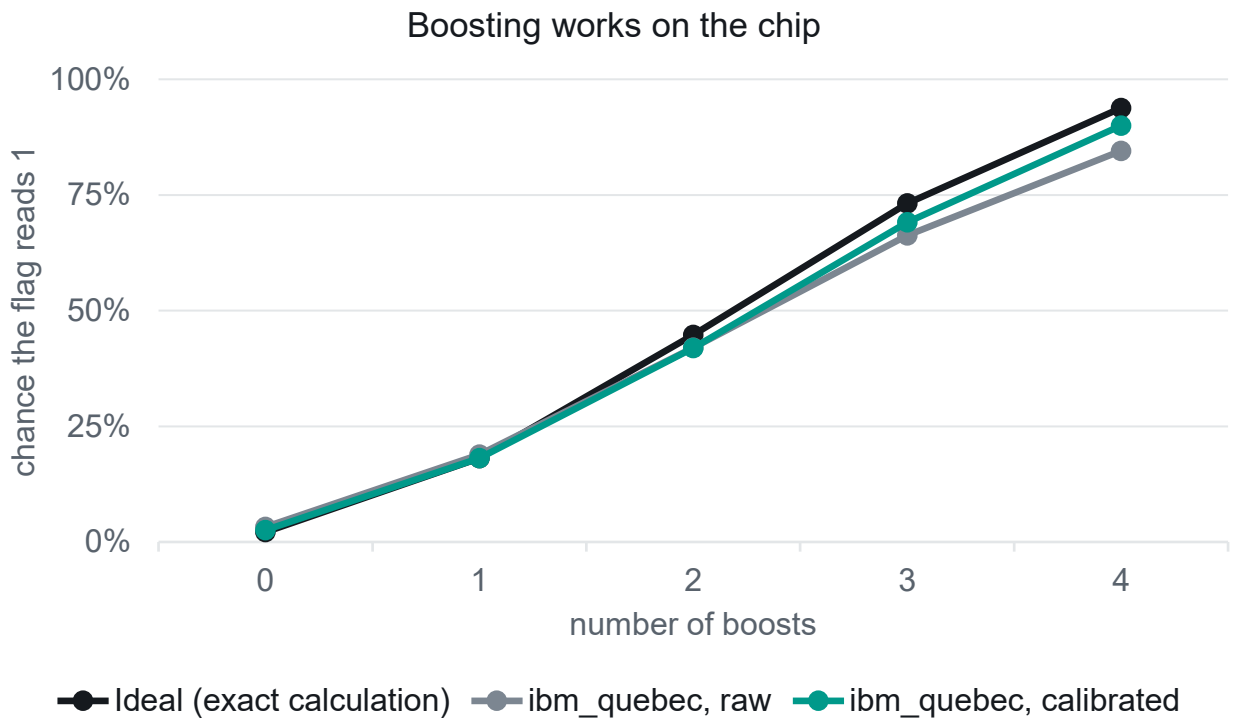
184 vs 34 of 200
confident right calls, hardest plan (9.51%)

One hour near the 2.8 h line: 24k runs (quantum), 82k (tuned importance sampling), 2.1M (Monte Carlo). Far from the line, importance sampling is cheaper.

experiments/coldsnap_plans.py: real 22-qubit circuits on Qiskit Aer (noise-free), 200 tries of 3,500 runs per check, against exact enumeration. Single-hour counts: experiments/turbine_decision.py, importance sampling tuned over 30 settings.

HARDWARE · IBM_QUEBEC

On a real IBM chip: it works, and it ties



How to read: each boost should raise the flag's chance along a known curve (black); fitting that curve gives p. Grey: the raw chip. Teal: after calibrating with two "twin" circuits whose answers we know.

WHAT WE RAN ON IBM QUANTUM HARDWARE

ibm_quebec 156-qubit IBM Heron chip	64 s of chip time, in 2 jobs
22 circuits 220,000 shots in total	3 qubits a real 2030 case, p = 2.13%

Works Calibrated, the chip follows the perfect curve: 90% after 4 boosts vs 94% ideal

Tie Error in p with 4,000 runs: chip 9.1%, Monte Carlo 10.7%. Within the noise

Loses Past about 30 two-qubit gates (2+ boosts), or counting the twin runs, Monte Carlo wins

Simulator, IBM noise model: a noise-aware fit beats Monte Carlo from 9,000 runs (2.8% vs 5.5% error at 16,000)

IBM Quantum ibm_quebec (156-qubit Heron), Qiskit IBM Runtime SamplerV2 with gate twirling and dynamical decoupling; jobs dbnk3tgqtr0c73ad7a9g and dbnkiblt7fqc73ajposg. Noise-aware fit: Tanaka et al. (2021). Noise model: Qiskit FakeFez.

What this does not show yet

LIMITS

- Supply vs demand only: no power-line limits, reservoir levels or imports from Nova Scotia
- Four winters of demand data, scaled to Hydro's peak forecasts
- Wind from weather data, no icing losses: optimistic
- Quantum savings assume a perfect, error-free quantum computer

WHAT DIDN'T WORK

- A quantum search for the best of the 4,096 plans: slower than a simple search in 41 of 45 tests
- Today's chip: noise wins after about 30 two-qubit gates
- No model predicts the Feb 2023 flash-freeze demand record

NEXT STEPS

- Add power-line limits and Maritime Link imports
- Bigger cold snaps (more units, more days), where only simulation works
- Short chip runs (1 to 2 boosts) with better error handling
- Take the ~3% break-even to the utility board (PUB)

Resources: a laptop CPU (45 min to rerun everything), Qiskit Aer, 64 s on ibm_quebec. Claims hold only for the tested model and sizes.

Layden et al. (2023), quantum-enhanced Markov chain Monte Carlo, Nature 619. Assumptions and costs are listed in the notebook (sections 13 and 14) and gridrisk/portfolio.py.

IN SHORT

Lights on, oil off, if the link holds

1.1%

The link was down 1.1% of the past year. At that rate, Unit 8, wind and heat pumps replace Holyrood's oil with no new diesel turbine

~3%

If the link is down more than about 3% of the time, the turbine is worth building. That is the number to debate

184 vs 34

Hardest cold-snap call, same runs: quantum right 184 times in 200, Monte Carlo 34 (simulator). On today's chip: a tie

Repository

github.com/SaiGaneshPS/qiskit-fall-2026-unsupervised-engineers

notebook with saved outputs, source, and an interactive demo

Questions?

References and acknowledgements

1. NL Hydro, 2024 Resource Adequacy Plan, Rev. 2. Filed with the NL Board of Commissioners of Public Utilities (PUB), Aug 2024.
2. NL Hydro, 2025 Near-Term Reliability Report. PUB, Nov 2025.
3. NL Hydro, Q3 2026 Rolling 12-Month Asset Performance report. PUB, Nov 2026.
4. NL Hydro, 2026-27 Winter Readiness Report, October update. PUB, Oct 2026.
5. NL Hydro, 2025 Island Interconnected System Load Forecast. PUB, Nov 2025.
6. NL Hydro, Avalon Combustion Turbine application, Rev. 1 and evidentiary update. PUB, Apr 2026.
7. NL Hydro, Bay d'Espoir Unit 8 early execution update. PUB, Oct 2026.
8. NL Hydro, Effective Load Carrying Capability study. PUB, Dec 2025.
9. NL Hydro, DarkNL briefing to the PUB, Jan 2014; January 2026 power warning notices.
10. Statistics Canada / CCEI, High-Frequency Electricity Data, NL island demand.
11. Hersbach et al. (2020), The ERA5 global reanalysis, QJRMS 146; accessed via Open-Meteo.
12. NL industrial facilities GHG data 2016-22 (gov.nl.ca); ECCC large-facility emissions data (2026).
13. Statistics Canada table 38-10-0286-01; NRCan heat pump guide; NEEP cold-climate heat pump specification v3.1.
14. Brassard, Høyer, Mosca and Tapp (2002), Quantum amplitude amplification and estimation, Contemp. Math. 305.
15. Suzuki et al. (2020), Amplitude estimation without phase estimation, Quantum Inf. Process. 19, 75.
16. Tanaka et al. (2021), Amplitude estimation via maximum likelihood on noisy quantum computer, Quantum Inf. Process. 20, 293.
17. Woerner and Egger (2019), Quantum risk analysis, npj Quantum Information 5, 15.
18. Egger et al. (2021), Credit risk analysis using quantum computers, IEEE Trans. Computers 70(12).
19. Layden et al. (2023), Quantum-enhanced Markov chain Monte Carlo, Nature 619, 282-287.
20. Qiskit 2.6, Qiskit Aer 0.18, Qiskit IBM Runtime; IBM Quantum ibm_quebec. scikit-learn (Pedregosa et al. 2011), NumPy, SciPy, pandas, Matplotlib.

Our work: the demand model, grid model, plan scoring, quantum circuits and every experiment. We reused the open-source libraries above and public data only. AI coding assistants helped with parts of the code and text; the team reviewed all of it and can explain every part.